

Siebert Relation Violation as a Probe of Spacetime Foam

A. G. Kim*

Physics Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

P. E. Nugent

*Applied Mathematics and Computational Research Division,
Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA*

L. Wang

Department of Physics and Astronomy, 4242 TAMU, College Station, Texas 77843-4242, USA

(Dated: May 20, 2026)

Quantum gravity predicts that spacetime fluctuates stochastically at the Planck scale — a “foam” that imprints a tiny achromatic arrival-time jitter on light from distant sources. We show that this jitter affects first- and second-order temporal coherence, $g^{(1)}$ and $g^{(2)}$, differently, producing a violation of the Siebert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ that otherwise holds for all Gaussian chaotic sources, when the foam fluctuates on timescales shorter than the optical coherence time. For extended sources, cross-ray jitter additionally suppresses the zero-lag bunching peak, leading to $g^{(2)}(0) < 2$, through a mechanism that depends only on the accumulated path-length variance, independent of the temporal dynamics of the foam. Strongly lensed sources — whether persistent or transient — combine cosmological path length for maximum foam-variance accumulation with magnification-boosted photon rates, and represent the highest-priority targets when accessible. These observables provide constraints that are statistically competitive with and complementary to existing imaging tests, though exquisite control of systematic errors is required to access the full range of physically viable foam models.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance $\sigma_\ell^2(D) \propto D^{2(1-\alpha)}\ell_P^{2\alpha}$; three benchmark values $\alpha = 1$ (conservative), $2/3$ (holographic), $1/2$ (random-walk) are conventional and are defined fully in Section II. Two complementary channels have been used to test foam models. The imaging channel exploits the stochastic aperture phase $\sigma_\phi = 2\pi\sigma_\ell/\lambda$, which degrades the PSF by $S \approx \exp(-\sigma_\phi^2)$; detections of near-diffraction-limited AGN images with HST and ground-based telescopes exclude $\alpha \lesssim 2/3$ for $D \sim 1\text{--}10$ Gpc under the assumption of short-range transverse foam correlations [7–10]; models with long-range correlations evade this bound [11]. A second channel, time-of-flight, constrains Lorentz-invariance-violating (LIV) energy-dependent dispersion from GRBs and blazar flares [12–14]; because geometric-optics foam jitter is achromatic ($\delta t = \Delta\ell/c$, frequency-independent), these analyses do not constrain stochastic foam and address a physically distinct quantum-gravity effect (a related GRB 221009A analysis claims consistency with holographic foam blurring [15], though without formal statistical significance). Together the two channels leave the holographic case $\alpha = 2/3$ unconstrained, and both share a structural limitation: the imaging test saturates once fringes are erased, while the time-of-flight test is blind to achromatic jitter. A foam model below the imaging threshold is invisible to both. Whether observables sensitive to sub-threshold phase variance exist — achromatic and distinguishable from plasma scintillation — is the question we address here.

We develop a framework based on the first- and second-order coherence functions $g^{(1)}(\tau)$ and $g^{(2)}(\tau)$ of light from astrophysical sources. The first-order coherence – the field–field autocorrelation, equivalently the fringe visibility

* agkim@lbl.gov

at delay τ – inherits a foam suppression that interpolates continuously between unity and the Lieu–Hillman fringe-erasure limit [7], recovering the Strehl and fringe-visibility tests as limiting cases of the same statistic. The second-order coherence – the intensity–intensity autocorrelation measured in a Hanbury Brown–Twiss experiment [16] – is to leading order blind to the carrier-phase modulation that suppresses $g^{(1)}$, but acquires a distinct signature from the differential timing jitter between photon arrivals. The two together violate the Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ [17] in a foam-specific way, providing a discriminant that does not require the intrinsic source visibility to be modeled. The Siegert violation requires the foam to fluctuate on timescales shorter than the optical coherence time; the zero-lag bunching suppression $\delta g^{(2)}$ for extended sources, by contrast, depends only on $\sigma_\ell^2(D)$ and is independent of the foam’s temporal dynamics. For spatially resolved sources, transverse foam coherence produces an additional, geometrically distinct effect: a suppression of the zero-lag HBT bunching peak that depends on the source size and transverse foam coherence length $r_\perp$.

Intuitively, $g^{(1)}$ measures the phase modulation of the electric field by the foam. The second-order coherence $g^{(2)}$ measures the differential arrival-time jitter between two photons. If both photons traverse identical foam fluctuations, their relative timing is unchanged and $g^{(2)}$ is unaffected. A change in $g^{(2)}$ requires the two photons to experience *different* foam: either they follow the same path but are separated in time by more than the foam coherence time τ_{foam} , so the foam has changed between their passages, or they travel different paths through spatially distinct foam regions. Photons emitted at the same time (zero-lag) from the same sky position have their relative timing unchanged while photons from different sky positions can have their relative timing changed, an effect not captured by first-order coherence that leads to Siegert violation. This is why $g^{(2)}$ probes foam temporal dynamics and transverse spatial coherence in ways that $g^{(1)}$ cannot.

Next-generation sensitivity is achievable via two complementary paths: large-aperture single telescopes measuring the HBT effect, and multi-telescope intensity interferometer arrays such as VERITAS [18, 19] — both increase collecting area and improve signal-to-noise for nearby resolved sources, while the temporal $g^{(1)}$ channel is accessible to high-resolution spectroscopy. Should signals be detected in both point-like and extended sources, their combination would jointly probe $\{\alpha, r_\perp\}$ and provide a cross-check against common instrumental systematics.

The remainder of the paper is organized as follows. Section II develops the model: the spatiotemporal covariance of the foam-induced path-length fluctuation field, the differential-jitter distribution at a single ray, and its generalization to ray pairs. Section III computes the foam suppression of first-order coherence, derives closed-form results for Lorentzian and Voigt line profiles, and connects it to the published Strehl and fringe-visibility tests. Section IV develops the second-order coherence, including the conditional Siegert relation and its narrow-band limit. Section V quantifies the resulting Siegert violation. Section VI extends the analysis to spatially resolved sources, derives the suppression of the zero-lag HBT bunching peak, and identifies distance scaling and achromaticity as discriminants against conventional propagation phase screens. Section VII assesses detectability via single-aperture HBT measurements, derives aperture and signal-to-noise scaling, and surveys candidate astrophysical targets. Section VIII summarizes the main results.

II. MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (1)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for virtually all known astrophysical sources) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta \ell_{\text{tot}}(t). \quad (2)$$

In the geometric-optics limit the path-length shift $\Delta \ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity is a key discriminant between foam and conventional propagation effects.

B. Spatial and Temporal Fluctuations of Spacetime Foam

Following Ng and collaborators [4–6] (see also Ref. [3] for a review of foam models and their observational constraints), the single-ray path-length variance obeys the power-law scaling

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (3)$$

where D is the proper distance to the source, α parametrizes the foam model ($\alpha = 1$: conservative; $\alpha = 2/3$: holographic; $\alpha = 1/2$: random walk), $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m is the Planck length, and A_α is a dimensionless coefficient of order unity. Equation (3) describes a photon traveling from a source at distance D that passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path. The exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature like steps in a random walk ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, constrained by an area-law bound on the number of independent degrees of freedom per Planck area; and conservative foam ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

Because foam is a property of spacetime rather than of any preferred direction, the same dimensional logic that governs longitudinal path-length fluctuations motivates the existence of a characteristic transverse coherence length $r_\perp$. However, the longitudinal and transverse cases are geometrically distinct: in the longitudinal case D counts the number of cells traversed along the ray, while in the transverse case b is a separation between rays that share cells along the line of sight — the number of shared cells depends on both b and D in a way that is not captured by a simple dimensional analogy. Accordingly, $r_\perp$ is treated as an independent free parameter in Section VI. We denote the normalized transverse coherence function by $\Phi(b/r_\perp)$, satisfying $\Phi(0) = 1$ and $\Phi(b/r_\perp) \rightarrow 0$ for $b \gg r_\perp$. The functional form of Φ depends on the specific foam model. For the pixellon model, Lee, Zurek & Chen [11] find that the transverse correlation structure involves long-range angular correlations (a Legendre polynomial series in aperture-plane angle) that do not reduce to a simple $\Phi(b/r_\perp)$ ansatz, but whose long-range character strongly suppresses image blurring relative to the uncorrelated case. In the present framework Φ is treated as a free function; a foam signal would manifest as an angular-size dependence of $\delta g^{(2)}$ (Section VI).

Turning to the temporal evolution of spacetime foam — an aspect not specified by Eq. (3) — quantum foam models posit that each Planck-scale cell along the line of sight fluctuates in time. Two photons traversing the same cell at coordinate times separated by τ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\text{foam}}$ but decorrelated for $\tau \gg \tau_{\text{foam}}$, where τ_{foam} is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_P$ cells along the line of sight. The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. If τ_{foam} is of order the Planck time, $\ell_P/c \approx 5 \times 10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\text{foam}} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all τ of interest and every C_α -dependent temporal observable vanishes: the finite-lag Siegert discrepancy $\mathcal{S}(\tau)$ is suppressed by $[1 - C_\alpha(\tau)] \rightarrow 0$ (Eq. (32)), and the single-ray $g_{\text{obs}}^{(1)}$ recovers its intrinsic value because the foam suppression factor $\mathcal{F}_{\text{foam}} = e^{-\sigma_\ell^2[1-C_\alpha]} \rightarrow 1$ (Eq. (23)). The extended-source zero-lag bunching suppression $\delta g^{(2)}(0)$ is the one exception: it depends only on the single-ray variance $\sigma_\ell^2(D)$ through σ_0^2 (Eq. (42)) and is therefore independent of C_α , remaining active even in the quasi-static limit. The two observables thus probe complementary aspects of the foam temporal dynamics. The full spatiotemporal covariance between two rays separated by transverse distance b takes the factorized form

$$\langle \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}) \Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}') \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right), \quad (4)$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (so $b = 0$):

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (5)$$

This product form — separating the total variance from the normalized spatiotemporal shape — is a structural assumption, not a consequence of Eq. (3), that defines a tractable class of foam models within which the analytic framework is exact. The parameters A_α , $C_\alpha(\tau; D)$, τ_{foam} , $r_\perp$, and Φ_{eff} parameterize distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and each governs a distinct aspect of the signal’s detectability. The factorization is consistent with phenomenologies in which the longitudinal path-length variance, the temporal decorrelation, and the transverse coherence arise from statistically

independent mechanisms — e.g. the simple cellular pictures of Refs. [4–6] where each Planck-scale cell contributes an independent random kick.

The factorization is *not* generic: the pixellon model of Lee, Zurek & Chen [11] has long-range angular correlations whose spatiotemporal covariance does not reduce cleanly to $\sigma_\ell^2(D) C_\alpha(\tau) \Phi(b/r_\perp)$; the framework here applies to that model only as a heuristic approximation in the regime where transverse correlations have decoupled.

Observationally, factorization predicts that the ratio $[g_{\text{obs}}^{(2)}(\tau) - 1]_{\text{extended}}/[g_{\text{obs}}^{(2)}(\tau) - 1]_{\text{point}}$ equals Φ_{eff} (Section VI) and is independent of both τ and D ; any dependence of this ratio on lag or source redshift would falsify the factorization ansatz and indicate that a non-factorizing model (e.g. pixellon) is required. The two principal observables developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ and the zero-lag bunching suppression $\delta g^{(2)}$ — both descend from the single scaling relation Eq. (3), with Sections III–V probing its longitudinal projection and Section VI its transverse projection.

C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t + \tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time t :

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \equiv \delta t(t + \tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}'). \quad (6)$$

For fixed D , stationarity implies the statistics depend only on τ and the transverse separation $b(\boldsymbol{\theta}, \boldsymbol{\theta}') = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. Expanding the square and applying Eq. (4):

$$\begin{aligned} \sigma_\Delta^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau; \boldsymbol{\theta}) - \Delta \ell_{\text{tot}}(t; \boldsymbol{\theta}')]^2 \rangle \\ &= \frac{1}{c^2} \left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2 C_\alpha(\tau) \Phi\left(\frac{b}{r_\perp}\right) \right] \\ &= \frac{2\sigma_\ell^2(D)}{c^2} \left[1 - C_\alpha(\tau; D) \Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right) \right]. \end{aligned} \quad (7)$$

For $b \gg r_\perp$, $\Phi \rightarrow 0$ and the variance becomes $2\sigma_\ell^2/c^2$ at all lags, including $\tau = 0$ — a new effect absent in the point-source case, and one that is independent of the temporal fluctuation structure of the foam (i.e. independent of C_α). In the dynamical limit $C_\alpha \rightarrow 0$ this result holds for all separations b , independently of Φ .

By the central limit theorem (CLT), $\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta}, \boldsymbol{\theta}'}(\tau) \sim \mathcal{N}(0, \sigma_\Delta^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}')), \quad (8)$$

where σ_Δ^2 is taken from Eq. (7).

The Gaussian approximation follows from the CLT: for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_P \gtrsim 10^{46}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For the cross-ray case the additional condition $b \gg r_\perp$ is assumed, and is satisfied in the fully-resolved limit $\Phi_{\text{eff}} \rightarrow 0$ adopted in Section VII.

III. FIRST-ORDER COHERENCE

In Sections III–V we restrict to a point source ($b = 0$), for which the cross-ray differential jitter of Eq. (6) reduces to the single-ray differential jitter

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t), \quad (9)$$

with variance obtained from Eq. (7) at $\Phi(0) = 1$:

$$\sigma_\Delta^2(\tau; D) = \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)]. \quad (10)$$

Although Sections III–V restrict to a point source for notational compactness, this single-ray variance applies equally to extended sources at finite lag $\tau \neq 0$: under the factorized covariance Eq. (4), the temporal factor $C_\alpha(\tau; D)$ is identical for all ray pairs, so cross-ray transverse structure enters only at $\tau = 0$ through Φ_{eff} (Section VI).

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (11)$$

Substituting Eq. (1) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (12)$$

Since δt is independent of E_{src} , averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (13)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

B. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}, \quad (14)$$

where τ_L is the coherence time. Evaluating Eq. (13) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = e^{\sigma_{\Delta}^2/2\tau_L^2} \cdot \frac{1}{2} \left[e^{-\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L - \tau/\sigma_{\Delta}}{\sqrt{2}}\right) + e^{\tau/\tau_L} \operatorname{erfc}\left(\frac{\sigma_{\Delta}/\tau_L + \tau/\sigma_{\Delta}}{\sqrt{2}}\right) \right]. \quad (15)$$

In the weak-jitter limit $\sigma_{\Delta} \ll \tau_L$, the Gaussian average of $e^{-|\tau - \Delta|/\tau_L}$ over $\Delta \sim \mathcal{N}(0, \sigma_{\Delta}^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g_{\text{obs}}^{(1)}(\tau) \approx e^{-|\tau|/\tau_L} \left[1 + \frac{\sigma_{\Delta}^2(\tau)}{2\tau_L^2} + \mathcal{O}(\sigma_{\Delta}^4) \right]. \quad (16)$$

Here $\sigma_{\Delta}(\tau)$ is τ -dependent through $C_{\alpha}(\tau)$, and the expansion is in $\sigma_{\Delta}(\tau)/\tau_L$ at fixed τ . The correction is positive because $e^{-|\tau|/\tau_L}$ is convex and Jensen's inequality applies. The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved exactly.

For a Voigt profile $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$ — a Lorentzian of width τ_L (from optically-thick electron scattering or BLR turbulent broadening) convolved with a Gaussian of width τ_G (from bulk Doppler motions) — the jitter kernel combines with the source Gaussian into an effective width and mean,

$$\tilde{\sigma}_1^2 = \frac{\sigma_{\Delta}^2 \tau_G^2}{\sigma_{\Delta}^2 + \tau_G^2}, \quad \mu_1(\tau) = \frac{\tau_G^2 \tau}{\sigma_{\Delta}^2 + \tau_G^2}, \quad (17)$$

giving, for $\tau > 0$:

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_{\Delta}} e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)} e^{\tilde{\sigma}_1^2/2\tau_L^2} \frac{1}{2} \left[e^{-\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L} \operatorname{erfc}\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) \right]. \quad (18)$$

The normalization $g_{\text{obs}}^{(1)}(0) = 1$ is preserved: recall $\sigma_{\Delta} = \sigma_{\Delta}(\tau)$ with $\sigma_{\Delta}(0) = 0$ (Eq. (10)), so $\tilde{\sigma}_1(0) = 0$, the exponential prefactors approach unity, and the erfc terms reduce to $\operatorname{erfc}(0) + \operatorname{erfc}(0) = 2$. In the limit $\tau_G \rightarrow \infty$ (pure Lorentzian), $\tilde{\sigma}_1 \rightarrow \sigma_{\Delta}$ and $\mu_1 \rightarrow \tau$, so Eq. (18) reduces to Eq. (15). In the limit $\tau_L \rightarrow \infty$ (pure Gaussian), the erfc terms combine to unity and $g_{\text{obs}}^{(1)}(\tau) = (\tilde{\sigma}_1/\sigma_{\Delta}) e^{-\tau^2/2(\sigma_{\Delta}^2 + \tau_G^2)}$: the observed coherence is a Gaussian whose width is broadened in quadrature by the jitter.

C. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, $E_{\text{src}}(t) = A(t) e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c \Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t - \delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation; the observed field takes the form

$$E_{\text{obs}}(t) \approx A(t) e^{-i\omega_0 t} e^{i\omega_0 \delta t(t)}. \quad (19)$$

(Corrections to the envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$; for $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$ this parameter is ≈ 0.28 , so the exact convolution Eq. (15) should be used instead. For $\alpha = 2/3$ the analogous parameter is $\sim 10^{-11}$ (Section VB), so Eq. (21) is exact to any practical precision in the holographic case.) Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0 ; since $\Delta(\tau)$ is Gaussian (Section IIC),

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \quad (20)$$

This holds when $\tau \gg \tau_{\text{foam}}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\text{foam}} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (10):

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \exp(-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]), \quad (21)$$

where we define the rms accumulated phase

$$\sigma_\phi^2 \equiv \left(\frac{2\pi}{\lambda_{\text{obs}}} \right)^2 \int_0^L [1 + z(s)]^2 \frac{d\sigma_\ell^2}{ds} ds, \quad (22)$$

integrating the foam differential variance $d\sigma_\ell^2/ds = A_\alpha 2(1 - \alpha) s^{1-2\alpha} \ell_P^{2\alpha}$ along the photon's local proper path of total length L . The $(1 + z)^2$ weighting reflects that a foam cell at redshift z imparts an observer-frame time delay $\delta t_{\text{obs}} = (1 + z) \delta \ell_{\text{local}}/c$, and hence a phase $\delta \phi = (2\pi/\lambda_{\text{obs}}) (1 + z) \delta \ell_{\text{local}}$, because the photon's wavelength at z is $\lambda_{\text{local}}(z) = \lambda_{\text{obs}}/(1 + z)$. Switching integration variable to comoving distance via $ds = dr_C/(1 + z)$ yields the equivalent form $\sigma_\phi^2 = (2\pi/\lambda_{\text{obs}})^2 \int_0^{D_C} (1 + z) (d\sigma_\ell^2/ds) dr_C$. For $z \ll 1$ both forms reduce to the flat-space limit $\sigma_\phi = 2\pi \sqrt{A_\alpha} D^{1-\alpha} \ell_P^\alpha/\lambda$. The extended-source observable inherits the same cosmological correction via the observed-frame single-ray time-delay rms $\sigma_{\tau, \text{obs}} \equiv \sigma_\ell(D) \sqrt{1 - \Phi_{\text{eff}}}/c$, so $\sigma_0^2 = 2\sigma_{\tau, \text{obs}}^2$ (Eq. (42)). We write the foam suppression factor as

$$\mathcal{F}_{\text{foam}}(\tau) \equiv e^{-\sigma_\phi^2 [1 - C_\alpha(\tau; D)]}, \quad (23)$$

so $g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot \mathcal{F}_{\text{foam}}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\text{foam}}(0) = 1$. At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2 [1 - C_\alpha(\infty; D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). The narrow-band condition $\sigma_\ell/(c\tau_c) \ll 1$ is compatible with the weak-jitter expansion of Eq. (16) — the former controls envelope displacement, the latter the carrier-phase suppression — and when neither applies the full convolution Eq. (13) must be evaluated.

D. Connection to Fringe Visibility and Strehl Ratio

The fringe visibility and Strehl ratio are spatial-coherence observables — they probe phase variations across an aperture or between two separated telescopes, described by $g_{\text{obs}}^{(1)}(B)$ — while $g_{\text{obs}}^{(1)}(\tau)$ derived above describes temporal coherence along a single line of sight. The two cases are directly analogous: the temporal decorrelation $C_\alpha(\tau; D)$, measuring how the foam changes between two observations separated by time τ , maps to the spatial decorrelation $\Phi(B/r_\perp)$ (Section II), measuring how correlated the foam fluctuations are at two aperture points separated by baseline B . In both cases the foam suppression has the same form, with C_α replaced by Φ :

$$\mathcal{F}_{\text{foam}}^{\text{spat}}(B) \equiv e^{-\sigma_\phi^2 [1 - \Phi(B/r_\perp)]}. \quad (24)$$

This mirrors $\mathcal{F}_{\text{foam}}(\tau)$ exactly: $\Phi(0) = 1$ gives no suppression at zero baseline, while $\Phi(B/r_\perp) \rightarrow 0$ for $B \gg r_\perp$ gives maximum suppression $e^{-\sigma_\phi^2}$ — just as $\mathcal{F}_{\text{foam}}(0) = 1$ and $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$.

The fringe visibility at baseline B is then

$$V(B) = V_{\text{src}}(B) \cdot \mathcal{F}_{\text{foam}}^{\text{spat}}(B) = V_{\text{src}}(B) \cdot e^{-\sigma_\phi^2 [1 - \Phi(B/r_\perp)]}, \quad (25)$$

where $V_{\text{src}}(B) = |g_{\text{src}}^{(1)}|$ evaluated at angular scale λ/B . In the limit $B \gg r_\perp$ the two aperture points sample independent foam realizations and the suppression reaches its maximum $e^{-\sigma_\phi^2}$. Lieu & Hillman [7] implicitly assumed this limit: their erasure condition $\Delta\phi \geq 2\pi$ with $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$ [4] is precisely $\sigma_\phi \geq 1$, the limit $\mathcal{F}_{\text{foam}}^{\text{spat}} \rightarrow 0$. Equation (25) extends their binary criterion to a continuous suppression curve whose shape encodes $\Phi(B/r_\perp)$ — in direct analogy to how $g_{\text{obs}}^{(1)}(\tau)$ encodes the temporal decorrelation $C_\alpha(\tau; D)$.

The Strehl ratio for a single aperture of diameter D_{tel} follows the same logic with $B \sim D_{\text{tel}}$: when $r_\perp \ll D_{\text{tel}}$, aperture-plane phases are independent draws from the single-ray distribution and $S = e^{-\sigma_\phi^2}$. [20] Both Strehl and fringe visibility therefore measure the same underlying $\sigma_\phi^2 = (2\pi\sigma_\ell/\lambda)^2$ in the large-baseline limit $B \gg r_\perp$, and the published constraints [4, 7–10] are unified within this framework.

IV. SECOND-ORDER COHERENCE

We derive the second-order coherence $g_{\text{obs}}^{(2)}(\tau)$ for a point source, continuing the framework of Section III. The result has the same structure as Eq. (13): $g_{\text{obs}}^{(2)} - 1$ is a convolution of $g_{\text{src}}^{(2)} - 1$ with the identical Gaussian jitter kernel. Because the intensity $|E_{\text{obs}}|^2$ is blind to the carrier phase $e^{i\omega_0\delta t}$, foam leaves $g^{(2)}$ nearly unchanged in the narrow-band limit while strongly suppressing $g^{(1)}$ — the asymmetry that produces the Siegert violation developed in Section V.

The same argument makes $g^{(2)}$ practically immune to atmospheric seeing under current observing conditions, in contrast to $g^{(1)}$. Atmospheric turbulence introduces path-length changes of order $\Delta\ell_{\text{atm}} \sim 0.1\text{--}1$ mm, far exceeding the optical wavelength; this randomizes the carrier phase and destroys fringe visibility without adaptive optics. However, $g^{(2)}$ is blind to this carrier-phase modulation, and the atmospheric coherence time $\tau_0 \sim 1$ ms far exceeds the HBT correlation window $\tau_L \sim \text{ns}$, so both photons in a coincidence traverse a frozen atmosphere and the differential atmospheric jitter is negligible. This advantage narrows as detector timing resolution approaches the atmospheric timing jitter $\Delta\ell_{\text{atm}}/c \sim 1\text{--}3$ ps; sub-ps detectors will require atmospheric monitoring or site-selection to control this systematic.

By the same argument as Eq. (13) in Section III: conditioned on the jitter realization, the intrinsic Siegert relation gives $g_{\text{obs}}^{(2)}(\tau | \delta t) - 1 = g_{\text{src}}^{(2)}(\tau - \Delta(\tau)) - 1$, and averaging over the Gaussian jitter distribution yields

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau)) [g_{\text{src}}^{(2)}(\tau - \Delta) - 1] d\Delta. \quad (26)$$

The Gaussian kernel requires $\tau \gg \tau_{\text{foam}}$; for $\tau \lesssim \tau_{\text{foam}}$ Eq. (26) should be regarded as approximate (Section II B).

The intensity $I_{\text{obs}}(t) = |E_{\text{obs}}(t)|^2$ is blind to the carrier phase $e^{i\omega_0\delta t(t)}$: only the envelope matters. In the narrow-band limit $\delta t \ll \tau_c$, the jitter does not appreciably shift the envelope, $A(t - \delta t) \approx A(t)$, so $I_{\text{obs}}(t) \approx |A(t)|^2 = I_{\text{src}}(t)$ to leading order in $\delta t/\tau_c$. Consequently:

$$g_{\text{obs}}^{(2)}(\tau) = g_{\text{src}}^{(2)}(\tau) \quad (\text{leading order in } \delta t/\tau_c), \quad (27)$$

with corrections of order $(\sigma_\ell/c\tau_c)^2$ from both envelope-timing and carrier-phase contributions to the intensity. For $\alpha = 1/2$ at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$: $\delta t_{\text{rms}}/\tau_c \approx 0.28$, so corrections are at the $\sim 8\%$ level and the exact convolution Eq. (26) should be used for precision work. For $\alpha = 2/3$ the expansion parameter $\sigma_\ell/(c\tau_c)$ is $\sim 10^{-11}$ and Eq. (27) is exact to any practical precision.

By contrast, $g^{(1)}$ is sensitive to the carrier phase because $\sigma_\phi = \omega_0\sigma_\ell/c$ can be large even when $\delta t \ll \tau_c$: for the same parameters, $\sigma_\phi \approx 280 \gg 1$.

V. SIEGERT RELATION VIOLATION

The Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ holds for any stationary complex Gaussian field and provides a powerful internal consistency check: once $g^{(1)}$ is known, the expected $g^{(2)}$ is fixed. As foreshadowed in Section IV, foam violates this relation because $g_{\text{obs}}^{(1)}$ is suppressed by the carrier-phase factor $\mathcal{F}_{\text{foam}}$ while $g_{\text{obs}}^{(2)}$ is not. This section derives and characterizes the Siegert discrepancy $\mathcal{S}(\tau) \equiv [g_{\text{obs}}^{(2)} - 1] - |g_{\text{obs}}^{(1)}|^2$.

A. General Bound

Define the Siegert discrepancy:

$$\mathcal{S}(\tau) \equiv \left[g_{\text{obs}}^{(2)}(\tau) - 1 \right] - \left| g_{\text{obs}}^{(1)}(\tau) \right|^2. \quad (28)$$

Let $X \equiv g_{\text{src}}^{(1)}(\tau - \Delta(\tau))$, a complex random variable. From Eqs. (13) and (26):

$$g_{\text{obs}}^{(1)}(\tau) = \langle X \rangle_{\delta t}, \quad g_{\text{obs}}^{(2)}(\tau) - 1 = \langle |X|^2 \rangle_{\delta t}. \quad (29)$$

Therefore

$$\mathcal{S}(\tau) = \langle |X|^2 \rangle - |\langle X \rangle|^2 \geq 0, \quad (30)$$

by the Cauchy-Schwarz inequality, with equality only when $\Delta(\tau)$ has zero variance (i.e., at $\tau = 0$ or in the absence of foam). The Siegert relation is violated for all $\tau \neq 0$ whenever foam is present. The Siegert violation is not a non-classical signature: the foam-modified field remains a classical stochastic process, and the classical inequalities $g^{(2)}(0) \geq g^{(2)}(\tau)$ and $g^{(2)}(0) > 1$ are preserved; it is only the Gaussian-statistics identity relating $g^{(2)}$ to $g^{(1)}$ that is broken, due to the stochastic phase accumulated along the propagation path.

B. Sensitivity Scale Decoupling

The violation arises because $g^{(1)}$ and $g^{(2)}$ respond to foam on different scales. The first-order coherence is controlled by $\sigma_\phi = 2\pi\sigma_\ell(D)/\lambda$, the rms accumulated phase along a single ray; $g^{(1)}$ is sensitive to it because the carrier phase $e^{i\omega_0\delta t}$ appears directly in the field correlation. The second-order coherence is controlled by $\sigma_\Delta(\tau)/\tau_c$, the rms differential jitter scaled by the coherence time; $g^{(2)}$ is sensitive to it because intensity correlations depend on how much the jitter *changes* between two detection events, not its absolute value. These two scales are related by

$$\frac{\sigma_\Delta(\tau)}{\tau_c} = \frac{\sqrt{2[1 - C_\alpha(\tau)]} \sigma_\ell(D)}{c\tau_c} = \sigma_\phi \cdot \frac{\Delta\lambda}{\lambda} \cdot \sqrt{2[1 - C_\alpha(\tau)]}, \quad (31)$$

using $\omega_0\tau_c = \lambda/\Delta\lambda$. The factor $\Delta\lambda/\lambda \ll 1$ means that even when $\sigma_\phi \gg 1$ (fringes erased), $\sigma_\Delta(\tau)/\tau_c$ can be small — so $g^{(2)}$ remains near its intrinsic value while $g^{(1)}$ is suppressed. This decoupling is the physical root of the Siegert violation. At the fiducial parameters $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$, and the dynamical limit $C_\alpha \rightarrow 0$, the random-walk model $\alpha = 1/2$ gives $\sigma_\phi = 280$ and $\sigma_\Delta(\tau)/\tau_c = 0.40$, while the holographic model $\alpha = 2/3$ gives $\sigma_\phi = 2.5 \times 10^{-8}$ and $\sigma_\Delta(\tau)/\tau_c = 3.6 \times 10^{-11}$. For quasi-static foam ($C_\alpha \approx 1$) the ratio $\sigma_\Delta(\tau)/\tau_c$ vanishes at all lags. The $\alpha = 1/2$ case is excluded by imaging under the standard assumption of short-range transverse foam correlations [10]; models with long-range correlations evade this bound [11], and $\alpha = 2/3$ (holographic) is the most sensitive surviving model.

C. Explicit Form and Properties in the Narrow-Band Limit

In the narrow-band limit, $g_{\text{obs}}^{(2)} \approx g_{\text{src}}^{(2)}$ (Eq. (27)), while $g_{\text{obs}}^{(1)}$ is suppressed by $\mathcal{F}_{\text{foam}}(\tau)$ (Eq. (21)). Using the intrinsic Siegert relation $g_{\text{src}}^{(2)}(\tau) - 1 = |g_{\text{src}}^{(1)}(\tau)|^2$, to leading order:

$$\mathcal{S}(\tau) \approx |g_{\text{src}}^{(1)}(\tau)|^2 \left(1 - e^{-2\sigma_\phi^2[1 - C_\alpha(\tau; D)]} \right). \quad (32)$$

In the weak-jitter limit $\sigma_\phi \ll 1$, $\mathcal{S} \approx 2\sigma_\phi^2[1 - C_\alpha] |g_{\text{src}}^{(1)}|^2$ — linear in σ_ϕ^2 and small. In the strong-jitter limit $\sigma_\phi \gg 1$, the exponential vanishes and $\mathcal{S} \approx |g_{\text{src}}^{(1)}(\tau)|^2$: $g_{\text{obs}}^{(2)} - 1$ retains its intrinsic value while $|g_{\text{obs}}^{(1)}|^2 \rightarrow 0$, producing a full violation. Quasi-static foam ($C_\alpha \approx 1$) gives $\mathcal{S} \approx 0$ at all lags — foam is invisible to all coherence observables in this limit. The discrepancy $\mathcal{S}(\tau)$ is constructed directly from the two measurable quantities $|g_{\text{obs}}^{(1)}(\tau)|$ and $g_{\text{obs}}^{(2)}(\tau) - 1$ (or equivalently $\mathcal{S}(\nu)$ in the spectral domain), and the modulation beyond the prefactor $|g_{\text{src}}^{(1)}(\tau)|^2$ is controlled solely by α , D , and $C_\alpha(\tau; D)$, independently of the unknown source spectrum. For a Lorentzian source, $|g_{\text{src}}^{(1)}(\tau_L)|^2 = e^{-2}$, and in the dynamical limit $C_\alpha \rightarrow 0$, Eq. (32) reduces to

$$\mathcal{S}(\tau_L) = e^{-2} \left(1 - e^{-2\sigma_\phi^2} \right), \quad (33)$$

the detection observable at lag τ_L (saturating at $e^{-2} \approx 0.135$ for $\sigma_\phi \gg 1$).

The evaluation lag $\tau = \tau_L$ in Eq. (33) is representative but not optimal: $\mathcal{S}(\tau)$ peaks at a smaller lag set by the competition between the source-coherence falloff $e^{-2\tau/\tau_L}$ and the foam-induced rise $1 - e^{-2\sigma_\phi^2[1-C_\alpha(\tau)]}$. Differentiating Eq. (32) for a Lorentzian source yields the implicit condition

$$-\sigma_\phi^2 C'_\alpha(\tau_{\max}) e^{-2\sigma_\phi^2[1-C_\alpha(\tau_{\max})]} = \frac{1}{\tau_L} \left(1 - e^{-2\sigma_\phi^2[1-C_\alpha(\tau_{\max})]}\right). \quad (34)$$

In the weak-jitter limit ($\sigma_\phi^2 \ll 1$) this reduces to the σ_ϕ -independent statement $-C'_\alpha(\tau_{\max})/[1 - C_\alpha(\tau_{\max})] = 2/\tau_L$. For the canonical exponential decorrelation $C_\alpha(\tau) = e^{-\tau/\tau_{\text{foam}}}$,

$$\tau_{\max} = \tau_{\text{foam}} \ln\left(1 + \frac{\tau_L}{2\tau_{\text{foam}}}\right), \quad (35)$$

which interpolates between $\tau_{\max} \approx \tau_{\text{foam}} \ln[\tau_L/(2\tau_{\text{foam}})]$ for fast foam ($\tau_{\text{foam}} \ll \tau_L$) and $\tau_{\max} \rightarrow \tau_L/2$ for slow foam ($\tau_{\text{foam}} \gg \tau_L$). In the strong-jitter / dynamical-foam limit ($\sigma_\phi^2 \gg 1$, $C_\alpha \rightarrow 0$ above τ_{foam}), $\mathcal{S}(\tau) \rightarrow e^{-2\tau/\tau_L}$ is monotonically decreasing and the peak migrates to $\tau \rightarrow \tau_{\text{foam}}^+$, where $\mathcal{S}_{\max} \rightarrow 1$ when $\tau_{\text{foam}} \ll \tau_L$.

VI. EXTENDED SOURCES: TRANSVERSE FOAM COHERENCE

We now generalize from a point source to an extended incoherent source. The first-order coherence $g_{\text{obs}}^{(1)}$ turns out to be unchanged from the point-source result, because $g^{(1)}$ probes only single-ray statistics. The second-order coherence $g_{\text{obs}}^{(2)}$, however, acquires new structure through cross-ray differential jitter between rays arriving from different sky directions, producing a suppression of the zero-lag bunching peak that directly probes the transverse foam coherence length $r_\perp$.

A. Field Model and Spatiotemporal Covariance

We generalize to an extended incoherent source whose angular brightness distribution is $I(\boldsymbol{\theta})$ (normalized to unity). The analysis assumes a point (single spatial mode) detector; a finite aperture would introduce Van Cittert–Zernike geometric phases that modify the angular coherence. The detected field is a weighted superposition over rays labelled by direction offset $\boldsymbol{\theta}$:

$$E_{\text{obs}}(t) = \int d\boldsymbol{\theta} I^{1/2}(\boldsymbol{\theta}) E_{\boldsymbol{\theta}}(t - \delta t(t; \boldsymbol{\theta})), \quad (36)$$

where the $E_{\boldsymbol{\theta}}$ are independent, identically distributed complex Gaussian processes (one per sky direction), and each ray carries its own foam jitter $\delta t(t; \boldsymbol{\theta})$. The spatiotemporal covariance of the path-length aggregate is given by Eq. (4) of Section II, with $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ the transverse separation between rays evaluated at the source end. For foam distributed uniformly along the line of sight, the ray separation varies as $z|\boldsymbol{\theta} - \boldsymbol{\theta}'|$; evaluating at the source plane $b = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ introduces an $O(1)$ geometric factor that is absorbed into the free parameter $r_\perp$.

The first-order coherence is unchanged from the point-source case: spatial incoherence between source directions ($\langle E_{\boldsymbol{\theta}}^* E_{\boldsymbol{\theta}'} \rangle \propto \delta(\boldsymbol{\theta} - \boldsymbol{\theta}')$) forces only the diagonal to contribute, and the angular integral over I gives unity, so

$$g_{\text{obs}}^{(1)}(\tau) = g_{\text{src}}^{(1)}(\tau) \cdot e^{-\sigma_\phi^2[1-C_\alpha(\tau; D)]}, \quad (37)$$

identical to Eq. (21). The transverse foam structure leaves no imprint on $g^{(1)}$.

B. Second-Order Coherence for an Extended Source

The unnormalized intensity correlator is $\langle I(t)I(t+\tau) \rangle = \langle E^*(t)E(t)E^*(t+\tau)E(t+\tau) \rangle$, where $E \equiv E_{\text{obs}}$ is the field of Eq. (36). Expanding the double angular integral and applying the Wick (moment) theorem to the four-point function of the zero-mean complex Gaussian field $E_{\boldsymbol{\theta}}$ produces two pairings (the $I^{1/2}$ weight factors enter the angular integral and are suppressed here for clarity):

$$\langle E_{\boldsymbol{\theta}_1}^* E_{\boldsymbol{\theta}_2} E_{\boldsymbol{\theta}_3}^* E_{\boldsymbol{\theta}_4} \rangle = \langle E_{\boldsymbol{\theta}_1}^* E_{\boldsymbol{\theta}_2} \rangle \langle E_{\boldsymbol{\theta}_3}^* E_{\boldsymbol{\theta}_4} \rangle + \langle E_{\boldsymbol{\theta}_1}^* E_{\boldsymbol{\theta}_4} \rangle \langle E_{\boldsymbol{\theta}_3}^* E_{\boldsymbol{\theta}_2} \rangle. \quad (38)$$

The first pairing contributes $\langle I(t) \rangle \langle I(t+\tau) \rangle$ (the disconnected term, which cancels when forming $g^{(2)} - 1$). The second pairing gives the connected contribution. Because different source directions are spatially incoherent, $\langle E_{\theta}^*(s) E_{\theta'}(s') \rangle \propto \delta(\theta - \theta')$, so the connected pairing requires $\theta_1 = \theta_4$ and $\theta_3 = \theta_2$. The four-fold angular integral therefore collapses to a double integral over ray pairs (θ, θ') , each pair contributing its own jitter-convolved $g_{\text{src}}^{(2)}$:

$$g_{\text{obs}}^{(2)}(\tau) - 1 = \iint d\theta d\theta' I(\theta) I(\theta') \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_{\Delta}^2(\tau; \theta, \theta')) [g_{\text{src}}^{(2)}(\tau - \Delta) - 1] d\Delta, \quad (39)$$

where $\sigma_{\Delta}^2(\tau; \theta, \theta')$ is from Eq. (7) (Section II C). The key new feature relative to the point-source case is that for $b \gg r_{\perp}$, $\Phi \rightarrow 0$ and $\sigma_{\Delta}^2 \rightarrow 2\sigma_{\ell}^2/c^2$ at all lags, including $\tau = 0$. In the limit $\Phi \equiv 1$ (all rays transversely coherent), every kernel reduces to the single-ray $\mathcal{N}(0, \sigma_{\Delta}^2(\tau))$ and Eq. (39) reduces to the point-source result Eq. (26).

C. Effective Transverse Coherence Factor

To reduce Eq. (39) to a tractable form, define the effective transverse coherence factor:

$$\Phi_{\text{eff}} \equiv \iint d\theta d\theta' I(\theta) I(\theta') \Phi\left(\frac{b(\theta, \theta')}{r_{\perp}}\right). \quad (40)$$

This replaces the ray-dependent differential variance with a single effective variance:

$$\sigma_{\tau, \text{eff}}^2(\tau) \equiv \frac{2\sigma_{\ell}^2(D)}{c^2} \left[1 - C_{\alpha}(\tau; D) \Phi_{\text{eff}}\right], \quad (41)$$

and allows Eq. (39) to be approximated by the simpler 1D convolution of Eq. (26) with σ_{Δ}^2 replaced by $\sigma_{\tau, \text{eff}}^2$. The two limits are: $\Phi_{\text{eff}} \rightarrow 1$ (point-like source or $r_{\perp} \gg D \Delta\theta$) recovers the single-ray result; $\Phi_{\text{eff}} \rightarrow 0$ (fully incoherent, $r_{\perp} \ll D \Delta\theta$) maximizes the effective variance. The approximation becomes exact in both limits. At intermediate angular sizes ($r_{\perp} \sim D \Delta\theta$), $\Phi(b/r_{\perp})$ varies across ray pairs and the Φ_{eff} reduction introduces errors that have not been quantified here; they must be assessed jointly with the CLT breakdown of Section II C, since both occur in the same regime $b \sim r_{\perp}$.

D. Extended-Source Siegert Discrepancy and Zero-Lag Bunching

The qualitatively new physics is at $\tau = 0$, where the cusp in $g_{\text{src}}^{(2)}$ converts cross-ray differential jitter into a first-order (linear-in- s) foam signature absent in the point-source case. At finite lag, the cross-ray correction is $O((\sigma_{\text{eff}}/\tau_c)^2)$ in the narrow-band limit and $\mathcal{S}(\tau \neq 0)$ reduces to the point-source result Eq. (32).

At $\tau = 0$, $g_{\text{obs}}^{(1)}(0) = 1$ exactly by the spatial-incoherence argument of Section VI, so $\mathcal{S}(0) = g_{\text{obs}}^{(2)}(0) - 2$. Evaluating the zero-lag effective variance with $C_{\alpha}(0; D) = 1$ gives

$$\sigma_0^2 \equiv \frac{2\sigma_{\ell}^2(D)}{c^2} (1 - \Phi_{\text{eff}}), \quad (42)$$

which is strictly positive whenever $\Phi_{\text{eff}} < 1$, so the ray-pair mixture reduces to a single Gaussian kernel of width σ_0 .

For a Lorentzian source, $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L}$, so the Siegert relation gives $g_{\text{src}}^{(2)}(\tau) - 1 = e^{-2|\tau|/\tau_L}$ (a cusp at $\tau = 0$ with e-folding width $\tau_L/2$). The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma_0^2)$ then becomes, setting $s \equiv \sigma_0/\tau_L$ and splitting on the sign of Δ (both halves contributing equally by symmetry of $|\Delta|$),

$$\langle e^{-2|\Delta|/\tau_L} \rangle = \frac{2}{\sqrt{2\pi} \sigma_0} \int_0^{\infty} e^{-2\Delta/\tau_L} e^{-\Delta^2/2\sigma_0^2} d\Delta = e^{2s^2} \text{erfc}(\sqrt{2}s), \quad (43)$$

which equals 1 as $s \rightarrow 0$ (no suppression) and 0 as $s \rightarrow \infty$ via the standard asymptotic $e^{x^2} \text{erfc}(x) \sim 1/(x\sqrt{\pi})$ (full suppression, $g_{\text{obs}}^{(2)}(0) \rightarrow 1$). So

$$g_{\text{obs}}^{(2)}(0) = 1 + e^{2s^2} \text{erfc}(\sqrt{2}s), \quad s \equiv \sigma_0/\tau_L, \quad (44)$$

and the zero-lag bunching peak is suppressed below its ideal value of 2 by

$$\delta g^{(2)} \equiv 2 - g_{\text{obs}}^{(2)}(0) = 1 - e^{2s^2} \text{erfc}(\sqrt{2}s), \quad (45)$$

In the small- s limit, $\delta g^{(2)} \approx 2\sqrt{2/\pi} s + O(s^2)$, so the suppression is linear in s for a Lorentzian source.

$$\mathcal{S}(0) = -\delta g^{(2)} \leq 0. \quad (46)$$

This sign is qualitatively distinct from the point-source result $\mathcal{S}(\tau \neq 0) \geq 0$ of Section V: a negative zero-lag Siegert discrepancy is therefore a sign-flip diagnostic that distinguishes spatially resolved from unresolved foam signatures. In the point-source limit $\Phi_{\text{eff}} \rightarrow 1$, $\sigma_0 \rightarrow 0$, $\delta g^{(2)} \rightarrow 0$, and $\mathcal{S}(0) \rightarrow 0$, recovering the point-source result. Since $\sigma_0^2 \propto (1 - \Phi_{\text{eff}})$ and Φ_{eff} decreases as the source angular diameter increases at fixed D and $r_\perp$, more extended sources experience greater suppression of the bunching peak and a more negative Siegert discrepancy at zero lag.

For an astrophysically more representative Voigt profile — a Lorentzian of width τ_L convolved with a Gaussian of width τ_G , where the Lorentzian component arises from optically-thick electron scattering in dense outflows (LBVs, Type IIIn supernovae, Population A NLS1) or from broad-line-region turbulent broadening [21], and the Gaussian component arises from bulk kinematics (Keplerian rotation, thermal Doppler, outflows) — the coherence function is $g_{\text{src}}^{(1)}(\tau) = e^{-|\tau|/\tau_L} e^{-\tau^2/2\tau_G^2}$, giving $g_{\text{src}}^{(2)}(\tau) - 1 = e^{-2|\tau|/\tau_L} e^{-\tau^2/\tau_G^2}$. The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma_0^2)$ combines the source Gaussian with the jitter kernel into an effective width

$$\tilde{\sigma}^2 = \frac{\sigma_0^2 \tau_G^2}{2\sigma_0^2 + \tau_G^2}, \quad (47)$$

yielding

$$g_{\text{obs}}^{(2)}(0) - 1 = \frac{\tilde{\sigma}}{\sigma_0} e^{2\tilde{s}^2} \text{erfc}(\sqrt{2}\tilde{s}), \quad \tilde{s} \equiv \tilde{\sigma}/\tau_L. \quad (48)$$

For any Voigt profile with finite τ_L , the parameter $\tilde{s} > 0$ and an $O(s)$ term is present, controlled by the Lorentzian component. A purely Gaussian source ($\tau_L \rightarrow \infty$, $\tilde{s} \rightarrow 0$) has no $O(s)$ correction: $g_{\text{obs}}^{(2)}(0) - 1 = 1/\sqrt{1 + 2s^2(\tau_c/\tau_G)^2}$.

In the narrow-band limit ($\sigma_\ell/(c\tau_c) \ll 1$, where τ_c is the post-filter coherence time, equal to τ_L when the filter passes the full Lorentzian line; Section III C), a Taylor expansion of the convolution integral recovers the small- s behaviour already visible in Eq. (45): sources with a smooth (even) profile have $g_{\text{src}}^{(2)'}(0) = 0$ and the correction is $O(\sigma_0^2)$, while for a Lorentzian the cusp $g_{\text{src}}^{(2)'}(0^+) = -2/\tau_L$ gives the linear term already noted. For a Voigt profile the expansion yields $\delta g^{(2)} \approx 2\sqrt{2/\pi} \sigma_0/\tau_L + (\sigma_0/\tau_G)^2 + \dots$: the $O(s)$ sensitivity is set entirely by τ_L , so spectrally resolved Lorentzian-component lines are the preferred targets.

For the region $\alpha > 1/2$ that experiments seek to probe, $\sigma_\phi \ll 1$ and the sensitivity parameter $s = \sigma_0/\tau_c = \sqrt{2(1 - \Phi_{\text{eff}})} \sigma_\phi \cdot (\Delta\lambda/\lambda)$ is small across all realistic observational bandwidths. For the holographic model ($\alpha = 2/3$), $\sigma_\phi \sim 10^{-8}$, so that even at a high spectral resolution $R = \lambda/\Delta\lambda \sim 10^4$, one finds $s \sim 10^{-12}$. In this regime the linear approximation $\delta g^{(2)} \approx 2\sqrt{2/\pi} s$ is essentially exact.

The suppression $\delta g^{(2)}$ arises from a purely spatial mechanism: photon pairs from directions $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ accumulate independent jitters when $b \gg r_\perp$, blurring $g^{(2)}(0)$ below 2, while $|g_{\text{obs}}^{(1)}(0)|^2 = 1$ exactly is unaffected — hence $\mathcal{S}(0) < 0$. This sign does not violate the single-ray bound $\mathcal{S} \geq 0$ of Section V, which is a variance for one random variable; for an extended source $g_{\text{obs}}^{(2)}(0) - 1$ is an angular integral over independent ray-pair kernels that admits no analogous lower bound.[22]

The coherence observables developed here serve as discriminants against conventional propagation phase screens [3], and for foam models with long-range transverse correlations — where image blurring is strongly suppressed [11] — they may be the only sensitive channel. The source-size dependence of $\delta g^{(2)}$ through Φ_{eff} provides a direct probe of $r_\perp$ with no analogue in a featureless propagation screen, which carries no transverse coherence structure. The mathematical structure of the foam suppression factor $\mathcal{F}_{\text{foam}}$ is identical to that of any Gaussian random phase screen, but two signatures distinguish foam: distance scaling and achromaticity. For spacetime foam $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ (D^1 for $\alpha = 1/2$, $D^{2/3}$ for $\alpha = 2/3$, D^0 for $\alpha = 1$), whereas a thin intervening screen scales as D and Kolmogorov turbulence as $D^{11/6}$; the holographic and conservative foam models predict sub-linear or flat growth qualitatively different from any known astrophysical screen. Geometrically, foam jitter ($\delta t = \Delta\ell/c$) is achromatic — independent of λ — in contrast to plasma-based effects which scale as λ^2 . The Fermi, H.E.S.S., and LHAASO time-of-flight analyses [12–14] rule out deterministic, energy-ordered LIV dispersion as a class of quantum-gravity effect, but do not constrain the stochastic achromatic foam jitter considered here (Section I); the principal chromatic alternative that the achromaticity discriminant addresses is therefore plasma propagation. In practice, the intrinsic source visibility $V_{\text{src}}(\tau)$ introduces a degeneracy with foam suppression, but the Siegert discrepancy $\mathcal{S}(\tau) = g_{\text{obs}}^{(2)}(\tau) - 1 - |g_{\text{obs}}^{(1)}(\tau)|^2$ is constructed directly from the data: a detection of $\mathcal{S} \neq 0$ requires no explicit model of the source spectrum, since $g_{\text{src}}^{(1)}(\tau)$ can be obtained from the source spectrum via the Wiener-Khinchin theorem.

VII. SIEGERT-VIOLATION: SENSITIVITY AND TARGET SELECTION

This section examines the detectability of Siegert violation via the HBT split-beam configuration [16, 23–25] and surveys candidate sources with Lorentzian (or Lorentzian-dominated) emission lines for which the sensitivity estimates apply directly. We consider two observables — temporal decorrelation at finite lag and transverse decorrelation at zero lag — and treat the resulting SNR figures as a statistical floor; in practice systematic errors will set the binding limit and a dedicated systematic budget is required for any claimed detection.

A. Sensitivity and Aperture Scaling

For a single-aperture HBT configuration using a 50/50 beamsplitter with unpolarized light [26], the per-lag noise on $g_{\text{obs}}^{(2)}$ is

$$\sigma[g_{\text{obs}}^{(2)}] = \frac{4}{R\sqrt{\tau_r T_{\text{obs}}}}, \quad (49)$$

where R is the total incident photon count rate, τ_r the timing resolution (correlation bin width) of the detectors, and T_{obs} the total observation time. The prefactor $4 = 2N_p$ with $N_p = 2$ polarization modes reflects that only photons sharing a polarization state contribute to the bunching signal, while the shot noise is set by the total (polarization-summed) count rate. The SNR for detecting a Siegert discrepancy of amplitude $\mathcal{S}(\tau)$ evaluated at a specific lag τ — treating $\mathcal{S}$ as constant within the bin of width τ_r — is then

$$\text{SNR}_{\mathcal{S}} = \frac{\mathcal{S}}{\sigma[g_{\text{obs}}^{(2)}]} = \frac{\mathcal{S} R}{4} \sqrt{\tau_r T_{\text{obs}}}. \quad (50)$$

All numerical projections adopt $A_\alpha = 1$ and the dynamical-foam limit $C_\alpha(\tau_L; D) \rightarrow 0$; the extended-source zero-lag $\delta g^{(2)}(0)$ is independent of C_α and survives in either limit.

The photon-counting rate is $R = \epsilon A \dot{n}$, where $\dot{n}$ is the source photon flux per unit area within the filter bandpass, ϵ the end-to-end throughput, and A the collecting area. We adopt the Kim *et al.* [27] baseline: $d = 9.96$ m ($A = 77.9$ m²), $\epsilon = 0.39$, and $\sigma_t = 13$ ps rms (30 ps FWHM). The astrophysical targets of interest have intrinsic coherence times $\tau_L \lesssim 1$ ps, so $\sigma_t \gg \tau_L$ for the baseline instrument and the measurement sits firmly in the detector-limited regime: an unresolved foam feature of width τ_L in a bin of width σ_t contributes an effective signal $(\tau_L/\sigma_t)|\mathcal{S}|$, so Eq. (50) becomes

$$\text{SNR}_{\text{det}} = \frac{|\mathcal{S}| R \tau_L}{4} \sqrt{\frac{T_{\text{obs}}}{\sigma_t}}. \quad (51)$$

We present the detector-limited SNR as the primary case throughout and quote the source-limited limit ($\tau_L \gtrsim \sigma_t$, requiring sub-ps timing) only as the gain that next-generation detectors would unlock.

Achieving a statistically significant SNR is a necessary but not sufficient condition for a foam detection. A separate and equally demanding challenge is controlling systematic errors—spurious correlations from detector afterpulsing, electronic crosstalk, and optical back-reflections produce positive artifacts near zero lag, atmospheric scintillation introduces classical intensity fluctuations that mimic $g^{(2)} > 1$, and detector dead time causes a compensating dip in $g^{(2)}$ near zero lag—to levels below the expected foam signal $|\mathcal{S}|$. Because the foam signatures are small ($|\mathcal{S}| \ll 1$ for all viable foam models), even modest systematics can mimic or mask the effect.

1. Temporal Decorrelation – Finite-lag variant

For a point-like Lorentzian source in the dynamical limit $C_\alpha(\tau_L; D) \rightarrow 0$, the Siegert discrepancy at $\tau \sim \tau_L$ is Eq. (33), $\mathcal{S}(\tau_L) = e^{-2}(1 - e^{-2\sigma_\phi^2})$ with σ_ϕ the rms accumulated phase (Eq. (22)): quadratic in σ_ϕ in the weak-foam limit ($\mathcal{S}(\tau_L) \approx 2e^{-2}\sigma_\phi^2$) and saturating at $e^{-2} \approx 0.135$ for $\sigma_\phi \gg 1$.

a. Detector-limited SNR. Substituting $\mathcal{S}(\tau_L)$ into Eq. (51),

$$\text{SNR}_{\text{det}}^{\text{pt}} = \frac{|\mathcal{S}(\tau_L)| R \tau_L}{4} \sqrt{\frac{T_{\text{obs}}}{\sigma_t}} \xrightarrow{\sigma_\phi \ll 1} \frac{1}{2} e^{-2} \sigma_\phi^2 R \frac{\tau_L}{\sqrt{\sigma_t}} \sqrt{T_{\text{obs}}}. \quad (52)$$

The detector-limited form carries a $\sqrt{\tau_L/\sigma_t}$ penalty relative to source-limited operation ($\text{SNR}_{\text{src}}^{\text{pt}} = \frac{1}{4} |\mathcal{S}(\tau_L)| R \sqrt{\tau_L T_{\text{obs}}}$); reaching the source-limited limit on broad-line quasars requires $\sigma_t \lesssim \tau_L \sim 0.1$ ps.

b. Filter strategy. The foam signal $\mathcal{S}(\tau_L)$ depends only on σ_ϕ , not on the filter width. The SNR (Eq. (52)) scales as $R\tau_L$, and these two quantities trade off against each other as the filter is narrowed: a narrower filter reduces the transmitted flux R but increases the post-filter coherence time $\tau_c = \lambda^2/(c\Delta\lambda_{\text{filter}})$. For a Lorentzian emission line, $R \propto \Delta\lambda_{\text{filter}}$ and $\tau_c \propto \Delta\lambda_{\text{filter}}^{-1}$ in the narrow-filter limit, so $R\tau_c$ is constant and the SNR is insensitive to filter width. The same cancellation holds for a flat continuum source. The practical lower bound on filter width is sky background: widening the filter beyond the emission line adds sky photons without increasing the source signal.

c. Distance scaling. Substituting $R \propto D^{-2}$ at fixed luminosity and $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ into Eq. (52) gives, in the weak-foam limit,

$$\text{SNR}^{\text{pt}} \propto D^{2(1-\alpha)} \cdot D^{-2} = D^{-2\alpha}, \quad (53)$$

saturating at $\text{SNR} \propto D^{-2}$ in the strong-foam limit. Since $\alpha > 0$, the SNR increases toward smaller D : nearby targets are preferred, provided the flat-space ($z \ll 1$) limit of Eq. (22) holds. At cosmological distances the D -dependence is modified by two competing effects. For a fixed rest-frame line, $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1+z)$ grows with redshift, reducing σ_ϕ via the $1/\lambda_{\text{obs}}^2$ prefactor in Eq. (22). Conversely, a fixed observer-frame bandpass accesses progressively bluer rest-frame transitions at higher redshift ($\lambda_{\text{rest}} = \lambda_{\text{obs}}/(1+z)$); since $\sigma_\phi \propto 1/\lambda_{\text{rest}}$, a source whose spectral line falls in the UV rest-frame accumulates substantially more phase than a lower-redshift counterpart observed at the same detector wavelength—potentially reversing the $D^{-2\alpha}$ preference.

The minimum detectable path-length amplitude follows from inverting Eq. (52) with the quadratic signal dependence $|\mathcal{S}(\tau_L)| \propto \sigma_\phi^2 \propto \sigma_\ell^2$:

$$\sigma_{\ell, \text{min}}^{\text{pt}} \propto \frac{\lambda}{d} \left(\frac{\sigma_t}{\tau_L^2 T_{\text{obs}}} \right)^{1/4}, \quad (54)$$

where $R \propto d^2$ has been used. Sensitivity improves as d^{-1} rather than d^{-2} , and as $T_{\text{obs}}^{-1/4}$ rather than $T_{\text{obs}}^{-1/2}$ — both costs of the quadratic dependence on σ_ϕ . Each additional decade of sensitivity in σ_ℓ requires four decades of integration time, compared with two for the extended-source variant.

2. Transverse Decorrelation – Zero-lag variant

For a spatially extended source whose angular size satisfies $D\Delta\theta \gtrsim r_\perp$ (equivalently $\Phi_{\text{eff}} \rightarrow 0$), photon pairs from different sky directions accumulate independent foam jitters and the zero-lag bunching peak is suppressed below 2 according to Eq. (44), with $s \equiv \sigma_0/\tau_L$ and $\sigma_0^2 = 2\sigma_\ell^2(D)(1 - \Phi_{\text{eff}})/c^2$ (Eq. (42)). The foam signal is the zero-lag Siegert discrepancy $\mathcal{S}(0) = g_{\text{obs}}^{(2)}(0) - 2 \leq 0$, a deficit relative to the Siegert prediction $g_{\text{src}}^{(2)}(0) = 2$. In the weak-foam limit $s \ll 1$ — the regime relevant to all viable foam models — $|\mathcal{S}(0)| = \delta g^{(2)} \approx 2\sqrt{2/\pi} s$ scales *linearly* in s , a consequence of the cusp in $g_{\text{src}}^{(2)}$ at $\tau = 0$ for Lorentzian-cored sources (Eq. (45), Section VID); the signal saturates at $|\mathcal{S}(0)| \rightarrow 1$ for $s \gg 1$.

Two structural conditions distinguish this variant from the finite-lag case and govern its applicability.

a. Source-size requirement. The cross-ray differential-jitter mechanism requires $D\Delta\theta \gtrsim r_\perp$ (equivalently $\Phi_{\text{eff}} < 1$); for sources unresolved on the foam transverse scale ($\Phi_{\text{eff}} \rightarrow 1$), $\sigma_0 \rightarrow 0$ and the variant becomes inactive, leaving the finite-lag variant of Section VII A 1 as the only HBT channel. Because $r_\perp$ is an unknown foam-model parameter, the source-size dependence of $\delta g^{(2)}$ across targets with varying $D\Delta\theta$ is the principal observational handle on $r_\perp$ (Section VI). The sensitivity analysis below adopts the fully-resolved limit $\Phi_{\text{eff}} = 0$ ($\sigma_0^2 = 2\sigma_\ell^2(D)/c^2$), giving the maximum signal each target can produce; for partially resolved targets the SNR scales by $\sqrt{1 - \Phi_{\text{eff}}}$.

b. Filter strategy: preserving the intrinsic cusp. The linear-in- s scaling of $\delta g^{(2)}$ requires that the filter pass the full emission line,

$$\Delta\lambda_{\text{filter}} \geq \text{FWHM}_{\text{line}}, \quad (55)$$

so that the post-filter coherence time is set by the intrinsic Lorentzian line width rather than the (smooth) filter profile; a narrower filter destroys the cusp in $g_{\text{src}}^{(2)}$ at zero lag and reverts $\delta g^{(2)}$ to $O(s^2)$.

c. Detector-limited SNR. Substituting $\mathcal{S}(0)$ into Eq. (51),

$$\text{SNR}_{\text{det}}^{\text{ext}} = \frac{|\mathcal{S}(0)| R \tau_L}{4} \sqrt{\frac{T_{\text{obs}}}{\sigma_t}} \xrightarrow{s \ll 1} \frac{1}{2} \sqrt{\frac{2}{\pi}} \sigma_0 R \sqrt{T_{\text{obs}}/\sigma_t}. \quad (56)$$

The SNR in this regime is independent of τ_L : the cusp signal $\propto 1/\tau_L$ is cancelled by the τ_L/σ_t dilution factor. The penalty relative to source-limited operation ($\text{SNR}_{\text{src}}^{\text{ext}} = \frac{1}{4} |\mathcal{S}(0)| R \sqrt{\tau_L T_{\text{obs}}}$, attainable only when $\sigma_t \lesssim \tau_L$) is $\sqrt{\tau_L/\sigma_t}$.

TABLE I. Candidate sources for the sensitivity analysis of Section VII. D is the luminosity distance (flat Λ CDM, Planck 2018 [28]); m is the apparent magnitude (band per source class); F is the integrated line flux; FWHM is observed-frame; $\tau_L = \lambda_{\text{obs}}^2 / (\pi c \Delta \lambda_{\text{FWHM}})$. Per-source caveats (sample-mean SN IIn scaling, η Car factor-of-a-few uncertainty, J1330–0905 image- Δ flux derivation) are given in Section VII B. [†]Adopted from the Kollatschny & Zetzl 2013 turbulent FWHM [21]; not resolved by the XP spectrum.

Source	z	D (Mpc)	m	Line	F ($\text{erg s}^{-1} \text{cm}^{-2}$)	FWHM ($\text{\AA}$)	τ_L (ps)
Mrk 766	0.013	59	13.6	H α	2.8×10^{-12} [29]	28 [21]	0.17
PHL 1811	0.192	971	14.4	H α	1.0×10^{-12} [30]	46 [30]	0.14
η Car (1843)	0	2.3×10^{-3} [31]	−1	H α	$\sim 5 \times 10^{-7}$ [32]	15 [32]	0.30
Gal. SN IIn	0	2×10^{-3}	−7.6	H α	$\sim 2 \times 10^{-4}$ [33, 34]	55 [33]	0.083
M31 SN IIn	0	0.78	5.3	H α	$\sim 1.4 \times 10^{-9}$ [33, 34]	55 [33]	0.083
J1330–0905 (A)	2.225	1.81×10^4	16.1	Ly α	5.1×10^{-13} [35]	52 [†]	0.031

d. Distance scaling. Substituting $R \propto D^{-2}$ at fixed luminosity and $\sigma_0 \propto \sigma_\ell(D) \propto D^{1-\alpha}$ into Eq. (56) gives, in the weak-foam limit,

$$\text{SNR}^{\text{ext}} \propto D^{1-\alpha} \cdot D^{-2} = D^{-(1+\alpha)}, \quad (57)$$

with the penalty steeper for larger α ($D^{-3/2}$ for $\alpha = 1/2$, $D^{-5/3}$ for $\alpha = 2/3$). In the strong-foam limit $|\mathcal{S}(0)| \rightarrow 1$ saturates and $\text{SNR} \propto D^{-2}$. A nearby source of modest luminosity but high apparent flux can therefore be competitive with a more luminous but distant source: the higher photon rate R at close range can compensate for the reduced foam signal $|\mathcal{S}(0)|$.

The minimum detectable path-length amplitude follows from inverting the weak-foam limit of Eq. (56) with $|\mathcal{S}(0)| \propto \sigma_0 \propto \sigma_\ell$:

$$\sigma_{\ell, \text{min}}^{\text{ext}} \propto \frac{\sqrt{\sigma_t}}{d^2 \sqrt{T_{\text{obs}}}}, \quad (58)$$

where $R \propto d^2$ has been used: sensitivity improves as $\sqrt{\sigma_t}$ in the detector-limited regime and scales as d^{-2} (collecting area) and as $T_{\text{obs}}^{-1/2}$, both more favourable than the corresponding d^{-1} and $T_{\text{obs}}^{-1/4}$ scalings of the finite-lag variant. Importantly, this variant is independent of the foam temporal correlation C_α : the cross-ray differential jitter that drives $\delta g^{(2)}(0)$ operates regardless of whether the foam is dynamically active or quasi-static, in contrast to the finite-lag variant which requires $\tau_{\text{foam}} \lesssim \tau_L$.

B. Candidate Sources

Two source properties drive target selection for both variants: the photon rate R , set by the total line flux captured within the filter, and the source distance D , which sets σ_ϕ (finite-lag, Eq. (52)) and σ_0 (zero-lag, Eq. (56)). For the zero-lag observable a third property matters: the line profile must retain a Lorentzian core to preserve the cusp in $g_{\text{src}}^{(2)}$ at zero lag and hence the linear-in- s scaling (Section VI D). The finite-lag observable imposes no line-profile constraint and is applicable to any spatially unresolved source. All sources discussed below have intrinsic coherence times $\tau_L \lesssim 1$ ps, well below the baseline timing resolution $\sigma_t = 13$ ps; the measurements therefore operate in the detector-limited regime throughout. The source classes surveyed below are illustrative rather than exhaustive; systematic searches through existing AGN flux catalogues and emission-line surveys will likely identify superior targets with higher photon rates, more purely Lorentzian line profiles, or more favorable distance-to-flux ratios. Observed properties and references for the principal candidates are collected in Table I.

Narrow-line Seyfert 1 galaxies as headline AGN targets. NLS1s (Population A; $\text{FWHM} \lesssim 4000 \text{ km s}^{-1}$; [36, 37]) have single-Lorentzian broad lines satisfying the cusp condition of Section VII B. We adopt two reference targets at complementary distances — the nearby Mrk 766 and the cosmological PHL 1811 (Table I; luminosity distances in flat Λ CDM with the Planck 2018 cosmology [28]); the H α and H β profiles of PHL 1811 are explicitly Lorentzian-fit by Leighly et al. 2007 (their §3) [30], providing direct empirical verification of the cusp condition for this target. A wide filter ($\Delta \lambda_{\text{filter}} \geq \text{FWHM}$) captures the full line flux, giving photon rates $R \approx 2.0 \times 10^5$ cps (Mrk 766) and $R \approx 8.5 \times 10^4$ cps (PHL 1811) on a 10 m telescope with $\epsilon = 0.39$; sky background and detector dark counts are negligible relative to the source rates, so the Brouri et al. correction for accidental coincidences is small [38]. For both targets the $2 \times \text{FWHM}_{\text{line}}$ filter width is narrow enough that the underlying pseudo-continuum (Fe II forest plus near-line continuum) dilutes the cusp signal by $\lesssim 10\%$ given the line-to-continuum ratio $\gtrsim 10:1$ typical of these objects;

the dilution does not change the qualitative SNR ordering but should be subtracted from the measured $g^{(2)}(\tau)$ as an additive constant when extracting $\delta g^{(2)}(0)$ in practice. The two targets span a factor of 16 in distance, exercising the $\sigma_\phi \propto D^{1-\alpha}$ scaling of the foam-vs-screen discriminant developed in Section VID.

Strongly-lensed quasars. Gravitational magnification by factors of ~ 10 –60 boosts the photon rate while preserving the cosmological foam-variance lever arm at the source’s intrinsic distance, making strongly-lensed quasars a complementary persistent-AGN target class. Persephone’s Torch (J1330–0905, $z = 2.22$, total magnification ~ 56 , brightest-image $i \approx 16$) [35] is included in Table I on the basis of its broad Ly α emission line measured directly from its GAIA DR3 XP spectrum (observed equivalent width 235 Å, integrated 3700–4100 Å line flux 1.28×10^{-12} erg s $^{-1}$ cm $^{-2}$ for the GAIA blob; image A carries a fraction 0.40 of the total system flux from the observed ratios $F_B/F_A = 0.74$, $F_C/F_A = 0.70$, $F_D/F_A = 0.08$, giving 5.1×10^{-13} erg s $^{-1}$ cm $^{-2}$ on image A). Under the Kollatschny & Zetzl 2013 [21] broad-line-region model the observed Ly α profile is a Voigt profile (Section VID, Eq. (48)): a Lorentzian core from BLR turbulent broadening ($\text{FWHM}_{\text{turb}} \sim 3800$ km s $^{-1}$) convolved with a rotational Gaussian, supplying the cusp condition for both observables; the Voigt reduction factor $\tilde{\sigma}/\sigma_0 = (1 + 2\sigma_0^2/\tau_G^2)^{-1/2} \approx 1$ in the weak-foam regime ($\sigma_0 \ll \tau_G$), so the rotational Gaussian envelope does not suppress the signal and the Lorentzian-core formula applies directly. The net finite-lag SNR exceeds PHL 1811 by $\sim 1.4\times$ at $\alpha = 2/3$ (Figure 1); the zero-lag channel is less suppressed because $\sigma_0 \propto \sigma_\ell/c$ depends on the path-length variance rather than λ_{obs} , and J1330–0905 is comparable to but slightly below the NLS1 baseline there (Figure 2). The Lorentzian-core hypothesis at high z is an extrapolation from the low- z K&Z sample; direct high-resolution spectroscopy of Ly α (or the redshifted near-infrared Balmer lines) would confirm it.

Galactic and Local-Group transients. Type IIIn supernovae and LBV giant eruptions — with η Car as the principal example — combine high H α luminosity with narrow Lorentzian line cores ($\tau_L \sim 0.1$ –0.3 ps), yielding SNR advantages of 10^4 – $10^6\times$ over the persistent-AGN baseline at Galactic distances (Section VII C). Three target-specific caveats apply. *Scattering geometry:* the Homunculus Nebula around η Car scatters substantial flux into the line of sight; the observed coherence is partially set by scattering geometry and must be modeled before crediting the photon count. *Stationarity:* the framework assumes a stationary Gaussian source, valid for integrations short relative to the variability timescale (weeks-to-months for LBV eruptions and SN IIIn light-curves); the 10 hr per-event integrations used in the figures satisfy this comfortably, while 10^4 hr campaign-scale totals require co-adding over stationary epochs (Section VII C).

Type IIIn supernovae. Rather than a specific event, we adopt a representative Type IIIn source characterized by the sample-mean peak H α luminosity $L(\text{H}\alpha) = 10^{41}$ erg s $^{-1}$ [33], broad-line FWHM ≈ 55 Å (H α rest-frame) [33], and apparent r -band magnitude from the sample-mean absolute magnitude $M_r = -19.16$ [34]. The H α flux at a given distance follows by inverse-square scaling; we evaluate two cases: a Galactic event at $D = 2$ kpc and an M31 event at $D = 778$ kpc (Table I).

C. SNR Results

SNR and signal amplitude as functions of the foam exponent α are shown in Fig. 1 (finite-lag variant) and Fig. 2 (zero-lag variant) for the six sources of Table I at the Kim et al. [27] baseline ($d = 9.96$ m, $\epsilon = 0.39$, $\sigma_t = 13$ ps rms) and $T_{\text{obs}} = 10$ hr per exposure (multiply by $\sqrt{10^3} \approx 31.6$ for a 10^4 hr campaign).

a. Strong-foam regime ($\alpha \lesssim 0.54$). At and below the Lieu–Hillman imaging exclusion both observables saturate: the finite-lag signal at $\mathcal{S}(\tau_L) \rightarrow e^{-2} \approx 0.135$ (Eq. (33)) and the zero-lag signal at $|\mathcal{S}(0)| \rightarrow 1$ (Eq. (45)), producing the flattening of every SNR curve visible in Figs. 1–2 at small α . On the plateau the SNR is set entirely by the photon rate R and is independent of α . For the highest- R targets the saturation level itself is detectable in $T_{\text{obs}} = 10$ hr: a Galactic SN IIIn at $\alpha = 0.54$ reaches SNR ~ 280 in the zero-lag channel and ~ 15 in the finite-lag channel, and an η Car-class eruption is at the threshold of detection (SNR ~ 0.7 extended, ~ 0.2 finite-lag). The HBT framework is therefore statistically competitive with — and methodologically complementary to — the imaging tests that set the $\alpha = 0.54$ boundary, with the advantage of sensitivity to foam models with long-range transverse correlations that evade those tests [11].

b. Holographic regime ($\alpha = 2/3$). All signals and SNRs lie many orders of magnitude below unity. On Mrk 766, $\sigma_\phi \sim 7.3 \times 10^{-9}$ gives $\mathcal{S}(\tau_L) \sim 1.5 \times 10^{-17}$ (finite-lag) and $\delta g^{(2)}(0) \sim 3.5 \times 10^{-11}$ (zero-lag) — a six-order-of-magnitude advantage of the zero-lag observable over the finite-lag one, mirrored in the SNRs ($\sim 1.6 \times 10^{-11}$ vs. $\sim 6.6 \times 10^{-18}$ at $T_{\text{obs}} = 10$ hr). The extended-source channel is therefore the holographic-regime channel of choice; the finite-lag channel retains value as the only HBT probe sensitive to τ_{foam} and to spatially unresolved sources, but is not competitive on raw amplitude. At $\alpha = 2/3$ a Galactic SN IIIn or LBV eruption returns a $\sim 10^6\times$ (zero-lag) and $\sim 10^4\times$ (finite-lag) SNR gain over the persistent-AGN baseline.

c. Persistent AGN: Mrk 766 and PHL 1811. The two NLS1 targets span a factor of 16 in distance and trade off photon rate against cosmological lever arm. PHL 1811’s $D^{1-\alpha}$ boost in σ_ℓ at $\alpha = 2/3$ approximately compensates its

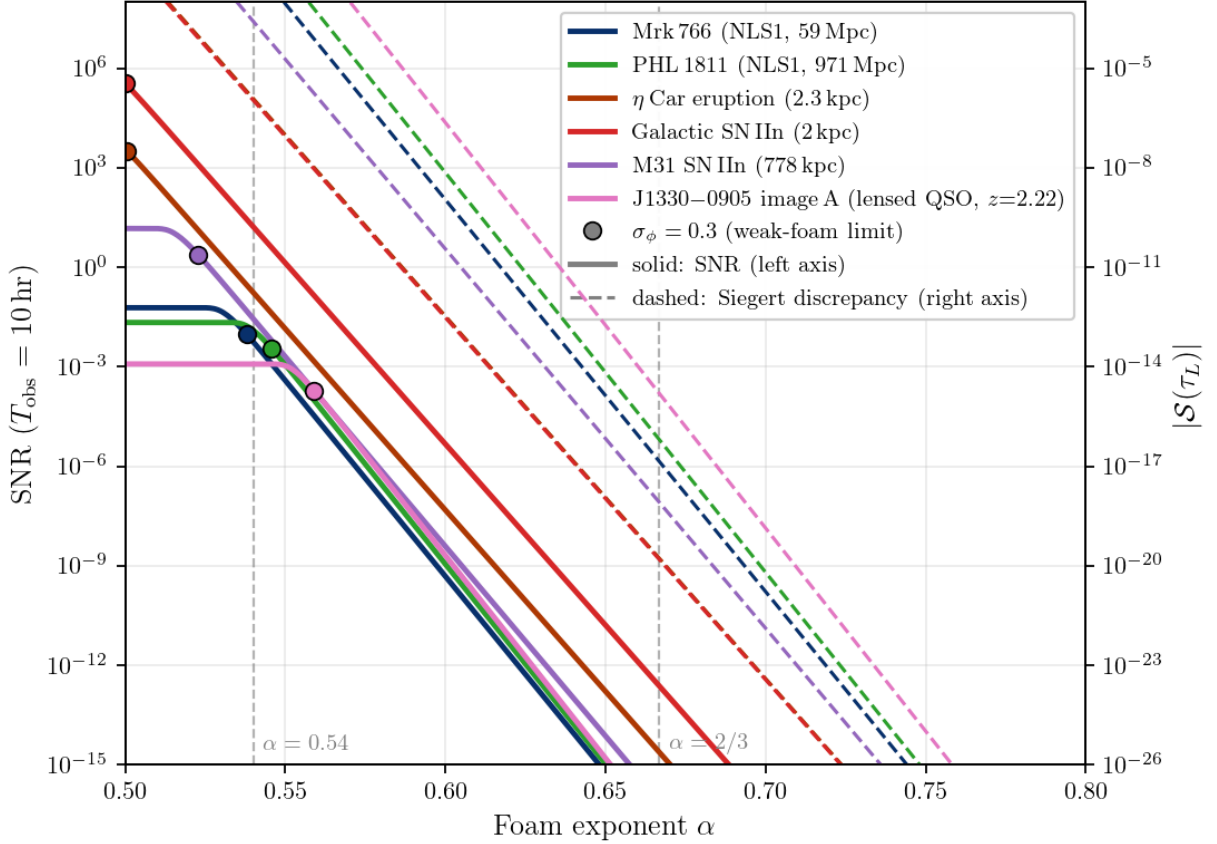


FIG. 1. Finite-lag Siebert discrepancy SNR for $\mathcal{S}(\tau_L) = e^{-2}(1 - e^{-2\sigma_\phi^2})$ (Eq. (33)) as a function of foam exponent α , in the dynamical-foam limit $C_\alpha \rightarrow 0$. Kim et al. [27] baseline (10 m, $\epsilon = 0.39$, $\sigma_t = 13$ ps rms), $T_{\text{obs}} = 10$ hr per exposure (multiply SNR by $\sqrt{10^3} \approx 31.6$ for the 10^4 hr campaign quoted in Sections VII C and VIII); detector-limited regime (Eq. (52)). Filled circles mark the $\sigma_\phi = 0.3$ weak-foam validity boundary; curves to the right saturate at $e^{-2} \approx 0.135$. Vertical dashed lines: $\alpha = 0.54$ (Lieu–Hillman imaging exclusion at $D = 1$ Gpc, $\lambda = 500$ nm) and $\alpha = 2/3$ (holographic). J1330–0905 is evaluated on the GAIA DR3 XP-measured Ly α flux with the Kollatschny & Zetzl 2013 turbulent-Lorentzian core (Section VII B). Dashed curves (right axis): $|\mathcal{S}(\tau_L)|$.

D^{-2} flux deficit, leaving the two within $\sim 10\%$ of each other in both channels (Mrk 766 nominally higher in the finite-lag channel, PHL 1811 marginally higher in the zero-lag channel; see Figs. 1–2). Their near-degeneracy makes the pair a useful internal consistency check on the $\sigma_\phi \propto D^{1-\alpha}$ scaling of the foam-vs-screen discriminant (Section VI D).

d. Lensed quasar J1330–0905: largest $|\mathcal{S}|$ of the persistent targets. The strongly-lensed quasar carries the cosmological lever arm to $z = 2.22$, giving $\sigma_\phi \approx 7.8 \times 10^{-8}$ on image A at $\alpha = 2/3$ — roughly $5\times$ that of PHL 1811. The resulting finite-lag signal $|\mathcal{S}(\tau_L)| \sim 1.7 \times 10^{-15}$ is $\sim 25\times$ that of PHL 1811 and the largest of the six sources; the zero-lag amplitude $|\mathcal{S}(0)| \sim 1.2 \times 10^{-9}$ exceeds PHL 1811’s by a factor of ~ 11 and is also the largest in the sample, despite the lower photon rate from a single magnified image (dashed curves, right axes of Figs. 1–2). For any HBT campaign that can absorb the photon-rate cost, J1330–0905 sets the high-water mark for foam signal amplitude among persistent targets.

e. Galactic and Local-Group transients. For a Galactic transient at $D \sim 2$ kpc with $\alpha = 1/2$ and long-range transverse correlations, the predicted $\sigma_\ell \approx 30$ nm and the photon flux is so high that detection or exclusion is essentially instantaneous once sub-ps timing is available. The Galactic SN IIn case yields the largest single-event SNR ($\sim 2 \times 10^6$ times the Mrk 766 baseline at $\alpha = 2/3$), but its recurrence rate makes it a “drop-everything” opportunity rather than a campaign target: Smith et al. 2011 [39] estimate that $\sim 9\%$ of core-collapse supernovae are Type IIn, and combined with the Galactic core-collapse rate of 1.6 ± 0.5 per century from Adams et al. 2013 [40] this implies a Galactic IIn rate of roughly one per 500–1500 yr; the most recent confirmed Galactic core-collapse supernova was Cassiopeia A (~ 1680 , Type IIb). The η Car case sacrifices a factor of ~ 10 in SNR but offers a recurrence plausibly within a working career. The M31 SN IIn is intermediate — plotted adjacent to its Galactic counterpart in Figs. 1–2

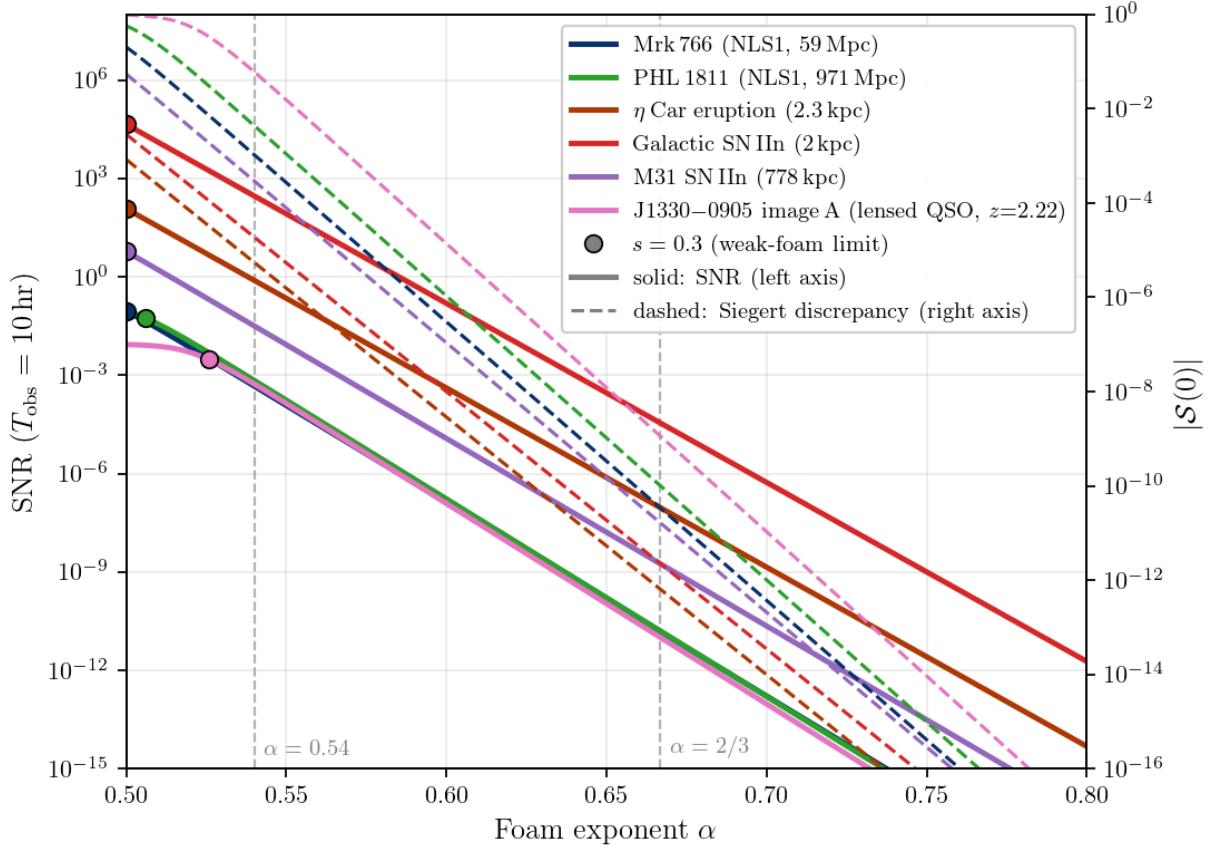


FIG. 2. Extended-source zero-lag bunching suppression $\delta g^{(2)}(0)$ SNR for $|S(0)| = 1 - e^{2s^2} \operatorname{erfc}(\sqrt{2}s)$ with $s = \sigma_0/\tau_L$ (Eq. (45)), as a function of foam exponent α , on the six representative targets. Instrument and integration as Figure 1; SNR from Eq. (56). All sources are in the fully-resolved limit $\Phi_{\text{eff}} = 0$ (signals scale by $\sqrt{1 - \Phi_{\text{eff}}}$ for partially resolved sources; Section VII A 2). Filled circles mark the $s = 0.3$ linear-approximation validity boundary. Vertical dashed lines: $\alpha = 0.54$ and $\alpha = 2/3$ (holographic). J1330–0905 (Section VII B) is included on the Kollatschny & Zetzl 2013 turbulent-Lorentzian basis ($\text{FWHM}_{\text{turb}} \sim 3800 \text{ km s}^{-1}$ for the $\text{Ly}\alpha + \text{N V}$ emitting region [21]). Dashed curves (right axis) show the corresponding Siebert discrepancy amplitude $|S(0)|$ for each source.

for direct comparison — and degrades the Galactic SNR by the inverse-square-distance factor $(778/2)^2 \approx 1.5 \times 10^5$. Its $\alpha = 1/2$ short- $r_{\perp}$ prediction ($\sigma_{\ell} \approx 625 \text{ nm}$) is already incompatible with existing imaging of M31, but the long- $r_{\perp}$ variant remains viable.

D. Sources Investigated but Not Competitive

The $\sigma_{\phi} \propto D^{1-\alpha}$ scaling means that any nearby source suffers a phase-variance penalty no realistic photon-rate or spectrograph multiplexing advantage can recover: at $D = 1 \text{ AU}$, σ_{ϕ}^2 is $\sim 10^{16}$ smaller than at $D = 1 \text{ Gpc}$ for $\alpha = 1/2$ and is utterly negligible for $\alpha = 2/3$. Three illustrative cases bear this out. *The Sun as a continuum source.* A $1 \text{ \AA}$ filter on a 10 m telescope yields a per-mode rate $r \sim 10^8 \text{ cps}$ but $\text{SNR}/\sqrt{T/\text{hr}} \sim 2 \times 10^{-4}$; even $N \sim 3000$ -channel spectrograph multiplexing reaches only $\sim 10^{-2}$ at $T_{\text{obs}} = 10^4 \text{ hr}$, far below detection. *Sirius B (best-case absorption-line target).* A broad absorption line can only *reduce* R relative to the unobscured continuum, so there is no photon-rate advantage from the absorption profile. The deep, Stark-broadened $\text{H}\alpha$ wing of this nearby DA white dwarf ($D = 2.64 \text{ pc}$) gives $\text{SNR}/\sqrt{T/\text{hr}} \approx 1.9 \times 10^{-3}$ at $\alpha = 1/2$ ($\sim 10\times$ below a Mrk 766-class NLS1) and is entirely negligible at $\alpha = 2/3$. Such targets remain useful for null-test calibration — a nearby DA white dwarf has a Lorentzian profile with negligible predicted foam signal for any plausible α . *The solar Ca II K line.* Combines proximity with extended-source dilution ($M \approx 4 \times 10^{10}$ independent spatial modes for a 10 m aperture); reaches $\text{SNR}/\sqrt{T/\text{hr}} \approx 2 \times 10^{-6}$, about $10^4\times$ below the AGN reference.

VIII. CONCLUSIONS

Two second-order coherence observables of spacetime foam emerge from this framework: a finite-lag Siegert violation $\mathcal{S}(\tau)$, requiring dynamically active foam, and a zero-lag bunching suppression $\delta g^{(2)}(0)$, requiring only $\sigma_\ell^2(D)$ and finite source angular size. Their value lies not in raw sensitivity — both signals are at $\sim 10^{-11}$ for the holographic $\alpha = 2/3$ model, well below current reach — but in three structural advantages over existing fringe-visibility and Strehl-ratio tests: (i) $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$ is constructed directly from the data, with $g_{\text{src}}^{(1)}(\tau)$ obtained from the source spectrum via Wiener–Khinchin, so a detection requires no separate source model; (ii) the cusp in $g_{\text{src}}^{(2)}$ at zero lag for Lorentzian-cored sources makes $\delta g^{(2)}$ scale linearly in $s = \sigma_0/\tau_L$ (Section VID), avoiding the double suppression suffered by first-order observables; (iii) the framework remains sensitive to foam models with long-range transverse correlations that evade imaging tests entirely [11], and supplies a chromatic discriminant against plasma scattering through the $\sigma_\phi \propto 1/\lambda$ scaling.

The framework identifies high-priority targets of opportunity: strongly lensed sources and nearby high-luminosity transients (Type II_n supernovae and LBV giant eruptions) that for $\alpha = 2/3$ offer $\sim 10^4$ – $10^6 \times$ the SNR of the persistent-AGN baseline (Section VII C). A standing rapid-response HBT plan — covering η Car, any galactic core-collapse supernova, and any nearby ($D \lesssim 30$ Mpc) Type II_n identified by transient surveys — would let the framework exploit these high-yield events without prior allocation. Strongly-lensed quasars (Section VII B; J1330–0905, comparable to the NLS1 baseline after cosmological correction) round out the persistent-AGN target class.

The achievable sensitivities are derived in Section VII. For a 10 m-class telescope [27] with $T_{\text{obs}} = 10^4$ hr on the Mrk 766-class NLS1 of Table I, the zero-lag extended-source channel (Eq. (58)) with current detectors ($\sigma_t = 13$ ps) reaches $\sigma_{\ell, \text{min}}^{\text{ext}} \approx 1.6 \mu\text{m}$ — above the Lieu–Hillman imaging floor of ~ 105 nm at H α . Next-generation sub-ps timing on a 40 m aperture pushes $\sigma_{\ell, \text{min}}^{\text{ext}}$ to ~ 9 nm, and a 1 km intensity-interferometer array to sub-nm, well below the imaging floor. What HBT-on-AGN cannot reach is the holographic prediction $\sigma_\phi \sim 10^{-8}$, six orders of magnitude below the most aggressive HBT projection and the natural target of tabletop holographic-noise interferometers [41]. Realising these projections demands parallel investment in systematic-error control: null-test calibration sources with negligible predicted foam signal are essential to verify that instrumental artifacts do not limit sensitivity before photon statistics do.

ACKNOWLEDGMENTS

This work was supported in part by the Director, Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.

The authors used Claude (Anthropic) and Valency Systems to assist with drafting and editing portions of this manuscript along with detailed reference searches. All content was reviewed and verified by the authors, who take full responsibility for the work.

We thank Robin Kaiser for helpful comments.

-
- [1] J. A. Wheeler, *Physical Review* **97**, 511 (1955).
 - [2] S. W. Hawking, *Nuclear Physics B* **144**, 349 (1978).
 - [3] S. Carlip, *Rep. Prog. Phys.* **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
 - [4] Y. J. Ng and E. S. Perlman, *Universe* **8**, 382 (2022).
 - [5] Y. J. Ng, *Physical Review Letters* **86**, 2946 (2001).
 - [6] Y. J. Ng, *Modern Physics Letters A* **18**, 1073 (2003).
 - [7] R. Lieu and L. W. Hillman, *The Astrophysical Journal* **585**, L77 (2003).
 - [8] R. Ragazzoni, M. Turatto, and W. Gaessler, *ApJ* **587**, L1 (2003), arXiv:astro-ph/0303043 [astro-ph].
 - [9] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, *Physical Review D* **83**, 084003 (2011).
 - [10] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, *The Astrophysical Journal* **805**, 10 (2015).
 - [11] V. S. H. Lee, K. M. Zurek, and Y. Chen, *Physical Review D* **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
 - [12] A. A. Abdo, M. Ackermann, M. Ajello, *et al.* (Fermi GBM/LAT Collaboration), *Nature* **462**, 331 (2009), arXiv:0908.1832 [astro-ph].
 - [13] F. Aharonian *et al.* (H.E.S.S. Collaboration), *Phys. Rev. Lett.* **101**, 170402 (2008), arXiv:0810.3475 [astro-ph].
 - [14] LHAASO Collaboration, *Physical Review Letters* **133**, 071501 (2024), arXiv:2402.06009.
 - [15] E. Steinbring and Y. J. Ng, arXiv e-prints (2026), arXiv:2603.18277.
 - [16] R. H. Brown and R. Q. Twiss, *Nature* **177**, 27 (1956).

- [17] A. J. F. Siegert, *On the fluctuations in signals returned by many independently moving scatterers*, Tech. Rep. 465 (MIT Radiation Laboratory, 1943).
- [18] A. U. Abeysekara *et al.*, *Nature Astronomy* **4**, 1164 (2020).
- [19] A. Acharyya *et al.*, *The Astrophysical Journal* **966**, 28 (2024).
- [20] The Maréchal formula $S = e^{-\sigma_\phi^2}$ (valid for $\sigma_\phi \lesssim 1$) and the temporal result $\mathcal{F}_{\text{foam}}(\infty) = e^{-\sigma_\phi^2}$ are linked by a spatio-temporal ergodic identification: both require $r_\perp \ll D_{\text{tel}}$ and $C_\alpha(\infty; D) \rightarrow 0$. For quasi-static foam ($C_\alpha \approx 1$) or $r_\perp \gg D_{\text{tel}}$, the aperture sits within a single coherence patch, the wavefront carries only a common phase offset, and $S \rightarrow 1$ even when σ_ϕ^2 is large. The transverse coherence function $\Phi(b/r_\perp)$ governs a physically distinct effect — cross-ray correlations at angular separations $\theta \neq \theta'$ — which enters the second-order coherence of extended sources (Section VI) but not the single-aperture Strehl ratio.
- [21] W. Kollatschny and M. Zetzl, *Astronomy & Astrophysics* **549**, A100 (2013).
- [22] Explicitly, $g_{\text{obs}}^{(2)}(0) - 1 = \int d\theta d\theta' I(\theta)I(\theta') \langle g_{\text{src}}^{(2)}(-\Delta_{\theta\theta'}) - 1 \rangle_{\delta t}$ (Eq. (39)), where each ray pair contributes with its own cross-ray variance $\sigma_\Delta^2(\tau; \theta, \theta')$; the Cauchy–Schwarz argument of Section V does not apply.
- [23] C. Foellmi, *Astronomy & Astrophysics* **507**, 1719 (2009).
- [24] D. Dravins, S. LeBohec, H. Jensen, and P. D. Nunēz, *Astroparticle Physics* **43**, 331 (2013).
- [25] W. Guérin, A. Dussaux, M. Fouché, G. Labeyrie, R. Donnerstein, D. Nakar, A. Gontier, and R. Kaiser, *Monthly Notices of the Royal Astronomical Society* **472**, 4126 (2017).
- [26] W. Guérin, M. Hugbart, S. Tolila, N. Matthews, O. Lai, J.-P. Rivet, G. Labeyrie, and R. Kaiser, *Comptes Rendus Physique* **26**, 659 (2025).
- [27] A. G. Kim, P. E. Nugent, X. Chen, L. Wang, J. T. O’Brien, and Tardis Collaboration, *Phys. Rev. D* **111**, 083047 (2025), arXiv:2503.07725 [astro-ph.IM].
- [28] Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, *et al.*, *Astronomy & Astrophysics* **641**, A6 (2020), arXiv:1807.06209.
- [29] A. Rodríguez-Ardila and X. Mazzalay, *Monthly Notices of the Royal Astronomical Society* **357**, 220 (2006).
- [30] K. M. Leighly, J. P. Halpern, E. B. Jenkins, and D. Casebeer, *Astrophysical Journal Supplement Series* **173**, 1 (2007), arXiv:0705.0940.
- [31] D. J. Hillier, K. Davidson, K. Ishibashi, and T. Gull, *The Astrophysical Journal* **553**, 837 (2001).
- [32] N. Smith, J. E. Andrews, A. Rest, F. B. Bianco, J. L. Prieto, T. Matheson, D. J. James, R. C. Smith, G. M. Strampelli, and A. Zenteno, *Monthly Notices of the Royal Astronomical Society* **480**, 1466 (2018).
- [33] F. Taddia, M. D. Stritzinger, J. Sollerman, M. M. Phillips, J. P. Anderson, L. Boldt, A. Campillay, S. Castellón, C. Contreras, G. Folatelli, S. M. Habergham, M. Hamuy, E. Heinrich-Josties, E. Y. Hsiao, W. Krzeminski, N. Morrell, C. R. Burns, W. L. Freedman, B. F. Madore, S. E. Persson, M. Roth, N. B. Suntzeff, P. A. James, and G. Folatelli, *Astronomy & Astrophysics* **555**, A10 (2013).
- [34] C. L. Ransome and V. A. Villar, *The Astrophysical Journal* **987**, 13 (2025), arXiv:2409.10596 [astro-ph.HE].
- [35] F. B. Davies, E. Bañados, S. E. I. Bosman, A. Ganguly, S. Belladitta, J. Power, and J. Rees, arXiv e-prints (2026), arXiv:2604.13152 [astro-ph.GA].
- [36] J. W. Sulentic, P. Marziani, R. Zamanov, R. Bachev, M. Calvani, and D. Dultzin-Hacyan, *The Astrophysical Journal* **566**, L71 (2002).
- [37] S. Zamfir, J. W. Sulentic, P. Marziani, and D. Dultzin, *Monthly Notices of the Royal Astronomical Society* **403**, 1759 (2010).
- [38] R. Brouri, A. Beveratos, J.-P. Poizat, and P. Grangier, *Optics Letters* **25**, 1294 (2000).
- [39] N. Smith, W. Li, A. V. Filippenko, and R. Chornock, *Monthly Notices of the Royal Astronomical Society* **412**, 1522 (2011), arXiv:1006.3899.
- [40] S. M. Adams, C. S. Kochanek, J. F. Beacom, M. R. Vagins, and K. Z. Stanek, *Astrophysical Journal* **778**, 164 (2013), arXiv:1306.0559.
- [41] E. P. Verlinde and K. M. Zurek, *Physics Letters B* **822**, 136663 (2021), arXiv:1902.08207 [gr-qc].